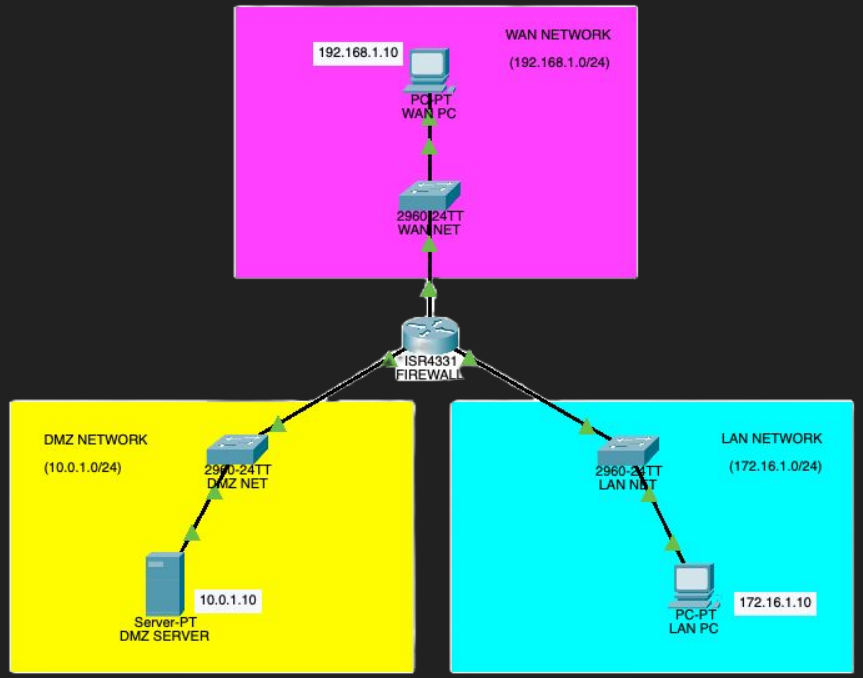
OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
DUBLIN
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Open Source Firewalls Assignment

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explanation of setup


WAN NETWORK
(192.168.1.0/24)
192.168.1.10
PC-PT
WAN PC
2960-24TT
WAN NET
ISR4331
FIREWALL
DMZ NETWORK
(10.0.1.0/24)
2960-24TT
DMZ NET
10.0.1.10
Server-PT
DMZ SERVER
LAN NETWORK
(172.16.1.0/24)
2960-24TT
LAN NET
172.16.1.10
PC-PT
LAN PC

Constraints

WAN to DMZ	DMZ to WAN
Allow HTTP 80 to server only	Internet access for updates (DNS, http, https)

LAN to DMZ	DMZ to LAN
Allow SSH from admin workstation and HTTP 80 for any device	Not allowed

LAN to firewall management	WAN/DMZ to firewall management
Allow from any device (anti-lockout)	Not allowed

LAN to WAN	WAN to LAN
Full access	Not allowed

overview of firewall configuration

Nftables Configuration (input)

```
# Input chain - traffic destined for the firewall
chain input {
    type filter hook input priority filter; policy drop;

    # Allow loopback traffic
    iif lo accept comment "Allow loopback"

    # Allow established and related connections
    ct state established,related accept comment "Allow established connections"

    # Block WAN direct access to firewall interfaces
    iif "enp0s3" ip daddr { 192.168.1.1, 10.0.1.1, 172.16.1.1 } counter log prefix "WAN-TO-FW-BLOCKED: " drop comment "Block WAN to firewall direct access"

    # Block ping from WAN to firewall
    iif "enp0s3" icmp type echo-request counter log prefix "WAN-PING-BLOCKED: " drop comment "Block WAN ping to firewall"

    # Allow ICMP for network diagnostics
    icmp type { echo-request, destination-unreachable, time-exceeded } limit rate 10/second accept
    icmpv6 type { echo-request, destination-unreachable, time-exceeded, nd-neighbor-solicit, nd-neighbor-advert } accept

    # Allow SSH from LAN to firewall
    iif "enp0s9" ip daddr 172.16.1.1 tcp dport 22 ct state new limit rate 5/minute accept comment "SSH from LAN with rate limiting"

    # Block SSH from DMZ to firewall
    iif "enp0s8" tcp dport 22 counter log prefix "DMZ-SSH-BLOCKED: " drop comment "Block DMZ SSH to firewall"

    # Block SSH from WAN to firewall
    iif "enp0s3" tcp dport 22 counter log prefix "WAN-SSH-BLOCKED: " drop comment "Block WAN SSH to firewall"

    # Allow DHCP if needed
    udp sport 68 udp dport 67 accept comment "DHCP client"

    # Allow NTP for time synchronization
    udp dport 123 accept comment "NTP"

    # Log and count dropped packets
    counter log prefix "INPUT-DROP: " drop
}
```

Nftables Configuration (forward)

```
    # Forward chain - traffic passing through the firewall
    chain forward {
        type filter hook forward priority filter; policy drop;

        # Allow established and related connections
        ct state established,related accept comment "Allow established connections"

        # WAN to DMZ - Allow web traffic to web server (with rate limiting, from WAN only)
        iif "enp0s3" oif "enp0s8" ip daddr 10.0.1.10 tcp dport @web_ports ct state new limit rate 20/second burst 40 packets accept comment "Web traffic to DMZ>

        # LAN to DMZ - Allow SSH with rate limiting (for administration)
        iif "enp0s9" oif "enp0s8" ip daddr 10.0.1.10 tcp dport 22 ct state new limit rate 3/minute accept comment "SSH to DMZ with rate limiting"

        # LAN to DMZ - Allow all traffic (internal users)
        iif "enp0s9" oif "enp0s8" tcp dport != 22 accept comment "LAN to DMZ access (no SSH)"

        # LAN to WAN - Allow internet access
        iif "enp0s9" oif "enp0s3" accept comment "LAN to internet"

        # DMZ to WAN - Limited outbound access for updates
        iif "enp0s8" oif "enp0s3" ip saddr 10.0.1.0/24 tcp dport { 80, 443, 53 } accept comment "DMZ outbound web/DNS"
        iif "enp0s8" oif "enp0s3" ip saddr 10.0.1.0/24 udp dport 53 accept comment "DMZ outbound DNS"
        iif "enp0s8" oif "enp0s3" ip saddr 10.0.1.0/24 udp dport 123 accept comment "DMZ NTP"

        # Block DMZ to LAN (critical security rule)
        iif "enp0s8" oif "enp0s9" counter log prefix "DMZ-to-LAN-BLOCKED: " drop comment "Block DMZ to LAN"

        # Block dangerous ports from WAN
        iif "enp0s3" tcp dport @blocked_ports counter log prefix "BLOCKED-PORT: " drop comment "Block dangerous ports"

        # Anti-spoofing rules
        iif "enp0s3" ip saddr { 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16, 127.0.0.0/8 } counter log prefix "SPOOFED-SRC: " drop comment "Block spoofed private>

        # Log and count all other drops
        counter log prefix "FORWARD-DROP: " drop
    }
```

Nftables Configuration (nat)

```
# NAT table for address translation
table ip nat {
    # Define web ports set in NAT table
    set web_ports {
        type inet_service
        elements = { 80, 443 }
    }

    # DNAT chain for incoming connections
    chain prerouting {
        type nat hook prerouting priority dstnat; policy accept;

        # Port forward web traffic to DMZ server
        iif "enp0s3" tcp dport @web_ports counter dnat to 10.0.1.10 comment "Forward web to DMZ"
    }

    # SNAT chain for outgoing connections
    chain postrouting {
        type nat hook postrouting priority srcnat; policy accept;

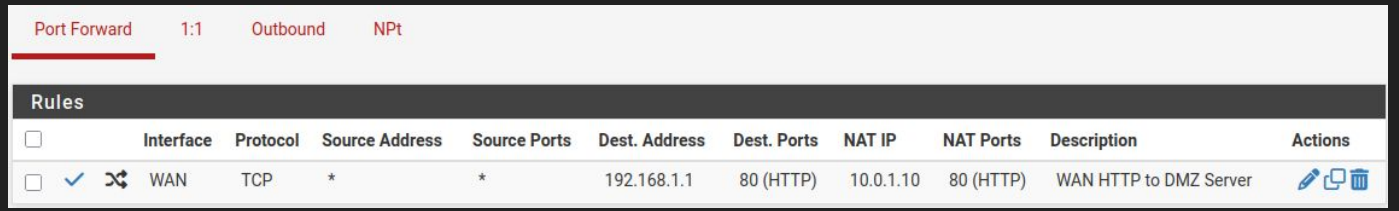
        # Masquerade LAN and DMZ traffic to WAN
        oif "enp0s3" ip saddr { 172.16.1.0/24, 10.0.1.0/24 } counter masquerade comment "NAT internal networks"
    }
}
```

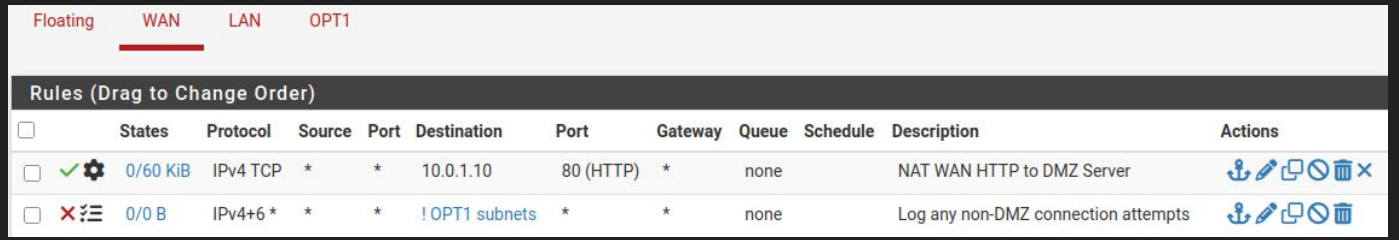
Nftables Configuration (logging)

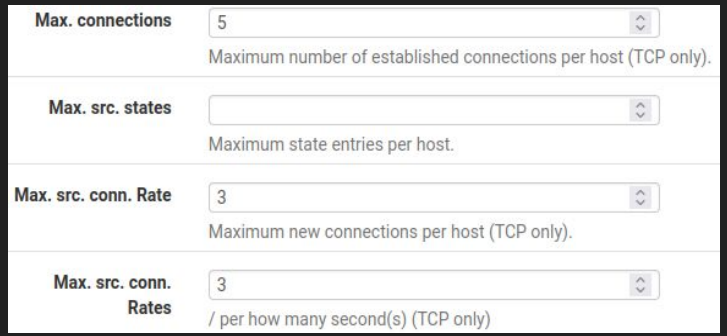
```
# Logging table for detailed monitoring
table inet logging {
    set suspicious_ips {
        type ipv4_addr
        size 1000
        flags timeout
        timeout 1h
    }

    chain log_suspicious {
        # Add IPs that exceed connection limits
        add @suspicious_ips { ip saddr limit rate over 100/minute }
        ip saddr @suspicious_ips counter log prefix "SUSPICIOUS-IP: " drop
    }
}
```

pfSense and OPNsense configurations (port forwarding and WAN)









pfSense and OPNsense configurations (LAN)

- Anti-lockout to firewall management
- DMZ Server admin management and partial access
- Internet access

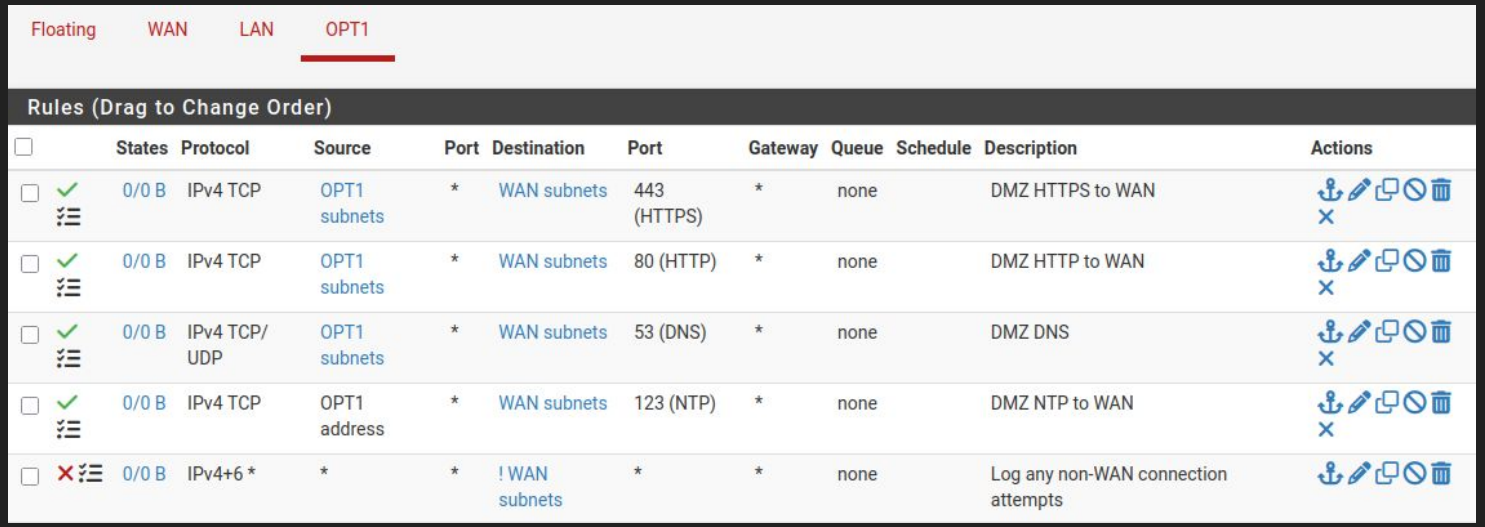
Floating WAN LAN OPT1

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	1/153 KiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	⚙
✓ ⚙	0/0 B	IPv4 TCP	172.16.1.10	*	OPT1 subnets	22 (SSH)	*	none		LAN admin workstation SSH to DMZ	⚓ ✎ ⧉ ⊘ 🗑 ✕
✓	0/0 B	IPv4 ICMP echoreq	LAN subnets	*	OPT1 subnets	*	*	none		LAN ping to DMZ	⚓ ✎ ⧉ ⊘ 🗑 ✕
✓	0/0 B	IPv4 TCP	LAN subnets	*	OPT1 subnets	80 (HTTP)	*	none		LAN HTTP to DMZ Server	⚓ ✎ ⧉ ⊘ 🗑 ✕
✓	0/0 B	IPv4 *	LAN subnets	*	WAN subnets	*	*	none		LAN Internet access IPv4	⚓ ✎ ⧉ ⊘ 🗑 ✕
✓	0/0 B	IPv6 *	LAN subnets	*	WAN subnets	*	*	none		LAN Internet access IPv6	⚓ ✎ ⧉ ⊘ 🗑 ✕

pfSense and OPNsense configurations (LAN)

- Partial WAN access for updates
- Logging of any other access attempts



Floating | WAN | LAN | OPT1

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/0 B	IPv4 TCP	OPT1 subnets	*	WAN subnets	443 (HTTPS)	*	none		DMZ HTTPS to WAN	
✓	0/0 B	IPv4 TCP	OPT1 subnets	*	WAN subnets	80 (HTTP)	*	none		DMZ HTTP to WAN	
✓	0/0 B	IPv4 TCP/ UDP	OPT1 subnets	*	WAN subnets	53 (DNS)	*	none		DMZ DNS	
✓	0/0 B	IPv4 TCP	OPT1 address	*	WAN subnets	123 (NTP)	*	none		DMZ NTP to WAN	
✗	0/0 B	IPv4+6 *	*	*	! WAN subnets	*	*	none		Log any non-WAN connection attempts	

relevant testing and analysis of results

Zone Isolation Testing

Test 1: LAN → DMZ (should succeed)

```
user@user:~$ ping -c 3 10.0.1.10
PING 10.0.1.10 (10.0.1.10) 56(84) bytes of data.
64 bytes from 10.0.1.10: icmp_seq=1 ttl=63 time=39.1 ms
64 bytes from 10.0.1.10: icmp_seq=2 ttl=63 time=15.3 ms
64 bytes from 10.0.1.10: icmp_seq=3 ttl=63 time=9.20 ms

--- 10.0.1.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2119ms
```

```
user@user:~$ curl -v 10.0.1.10
*   Trying 10.0.1.10:80...
* Connected to 10.0.1.10 (10.0.1.10) port 80
> GET / HTTP/1.1
> Host: 10.0.1.10
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Date: Thu, 26 Feb 2026 00:55:36 GMT
< Server: Apache/2.4.58 (Ubuntu)
< Last-Modified: Tue, 24 Feb 2026 17:11:04 GMT
< ETag: "135-64b94f8f8c15d"
< Accept-Ranges: bytes
< Content-Length: 309
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html>
<head>
    <title>DMZ Web Server</title>
</head>
<body>
    <h1>DMZ Web Server Test Page</h1>
    <p>Server: dmz</p>
    <p>IP Address: 10.0.1.10</p>
    <p>Timestamp: Tue Feb 24 04:56:36 PM GMT 2026</p>
    <hr>
    <p>This server is protected by nftables firewall</p>
</body>
</html>
* Connection #0 to host 10.0.1.10 left intact
```

```
user@user:~$ ssh user@10.0.1.10
user@10.0.1.10's password:
Last login: Wed Feb 25 18:40:15 2026 from 172.16.1.10
```

Zone Isolation Testing

Test 2: DMZ → LAN (should fail)

```
user@dmz:~$ ping -c 3 172.16.1.10
PING 172.16.1.10 (172.16.1.10) 56(84) bytes of data.

--- 172.16.1.10 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2063ms

user@dmz:~$ nc -zv 172.16.1.10 22
nc: connect to 172.16.1.10 port 22 (tcp) failed: Connection timed out
user@dmz:~$ nc -zv 172.16.1.10 80
nc: connect to 172.16.1.10 port 80 (tcp) failed: Connection timed out
user@dmz:~$
```

Zone Isolation Testing

Test 3: WAN → LAN (should fail)

```
┌──(kali㉿kali)-[~]
└─$ ping -c 3 172.16.1.10
PING 172.16.1.10 (172.16.1.10) 56(84) bytes of data.

--- 172.16.1.10 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2099ms
```

```
┌──(kali㉿kali)-[~]
└─$ nc -zv 172,16.1.10 22
172,16.1.10: forward host lookup failed: Unknown host
```

```
┌──(kali㉿kali)-[~]
└─$ telnet 172.16.1.10 80
Trying 172.16.1.10...
Connected to 172.16.1.10.
Escape character is '^]'.

HTTP/1.1 400 Bad Request
Date: Thu, 26 Feb 2026 03:05:03 GMT
Server: Apache/2.4.58 (Ubuntu)
Content-Length: 301
Connection: close
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>400 Bad Request</title>
</head><body>
<h1>Bad Request</h1>
<p>Your browser sent a request that this server could not understand.<br />
</p>
<hr>
<address>Apache/2.4.58 (Ubuntu) Server at 10.0.1.10 Port 80</address>
</body></html>
Connection closed by foreign host.
```

Zone Isolation Testing

Test 4: WAN → DMZ (should succeed)

only port 80

```
┌──(kali㉿kali)-[~]
└─$ curl -v http://192.168.1.1
*   Trying 192.168.1.1:80...
* Established connection to 192.168.1.1 (192.168.1.1 port 80) from 192.168.1.10 port 59286
* using HTTP/1.x
> GET / HTTP/1.1
> Host: 192.168.1.1
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Thu, 26 Feb 2026 02:39:03 GMT
< Server: Apache/2.4.58 (Ubuntu)
< Last-Modified: Tue, 24 Feb 2026 17:11:04 GMT
< ETag: "135-64b94f8f8c15d"
< Accept-Ranges: bytes
< Content-Length: 309
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html>
<head>
    <title>DMZ Web Server</title>
</head>
<body>
    <h1>DMZ Web Server Test Page</h1>
    <p>Server: dmz</p>
    <p>IP Address: 10.0.1.10</p>
    <p>Timestamp: Tue Feb 24 04:56:36 PM GMT 2026</p>
    <hr>
    <p>This server is protected by nftables firewall</p>
</body>
</html>
* Connection #0 to host 192.168.1.1:80 left intact
```

Service Access Control Testsing

Test 1: Authorized Port Access

```
┌──(kali㉿kali)-[~]
└─$ curl -v http://192.168.1.1:80
*   Trying 192.168.1.1:80...
* Established connection to 192.168.1.1 (192.168.1.1 port 80) from 192.168.1.10 port 54088
* using HTTP/1.x
> GET / HTTP/1.1
> Host: 192.168.1.1
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Thu, 26 Feb 2026 13:44:44 GMT
< Server: Apache/2.4.58 (Ubuntu)
< Last-Modified: Tue, 24 Feb 2026 17:11:04 GMT
< ETag: "135-64b94f8f8c15d"
< Accept-Ranges: bytes
< Content-Length: 309
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html>
<head>
    <title>DMZ Web Server</title>
</head>
<body>
    <h1>DMZ Web Server Test Page</h1>
    <p>Server: dmz</p>
    <p>IP Address: 10.0.1.10</p>
    <p>Timestamp: Tue Feb 24 04:56:36 PM GMT 2026</p>
    <hr>
    <p>This server is protected by nftables firewall</p>
</body>
</html>
* Connection #0 to host 192.168.1.1:80 left intact
```

WAN → DMZ (http – allow)

```
┌──(kali㉿kali)-[~]
└─$ ssh user@192.168.1.1
ssh: connect to host 192.168.1.1 port 22: Connection timed out
```

WAN → DMZ (ssh – block)

```
┌──(kali㉿kali)-[~]
└─$ nmap -p 22,80,443 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 08:48 -050
mass_dns: warning: Unable to determine any DNS servers. Reverse [
Nmap scan report for 192.168.1.1
Host is up (0.058s latency).

PORT    STATE    SERVICE
22/tcp  filtered ssh
80/tcp  open     http
443/tcp closed   https
MAC Address: 08:00:27:2F:7C:6C (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.91 seconds
```

Service Access Control Testsing

Test 2: Blocked Dangerous Ports WAN → DMZ (block all ports that are not 80)

```
┌──(kali㉿kali)-[~]
└─$ nc -zv 192.168.1.1 22
192.168.1.1: inverse host lookup failed: Host name lookup failure
(UNKNOWN) [192.168.1.1] 22 (ssh) : Connection timed out
```

```
┌──(kali㉿kali)-[~]
└─$ nc -zv 192.168.1.1 25
192.168.1.1: inverse host lookup failed: Host name lookup failure
(UNKNOWN) [192.168.1.1] 25 (smtp) : Connection timed out
```

```
┌──(kali㉿kali)-[~]
└─$ nc -zv 192.168.1.1 23
192.168.1.1: inverse host lookup failed: Host name lookup failure
(UNKNOWN) [192.168.1.1] 23 (telnet) : Connection timed out
```

```
┌──(kali㉿kali)-[~]
└─$ nc -zv 192.168.1.1 139
192.168.1.1: inverse host lookup failed: Host name lookup failure
(UNKNOWN) [192.168.1.1] 139 (netbios-ssn) : Connection timed out
```

```
┌──(kali㉿kali)-[~]
└─$ nc -zv 192.168.1.1 445
192.168.1.1: inverse host lookup failed: Host name lookup failure
(UNKNOWN) [192.168.1.1] 445 (microsoft-ds) : Connection timed out
```

```
┌──(kali㉿kali)-[~]
└─$ nc -zv 192.168.1.1 3389
192.168.1.1: inverse host lookup failed: Host name lookup failure
(UNKNOWN) [192.168.1.1] 3389 (ms-wbt-server) : Connection timed out
```

```
┌──(kali㉿kali)-[~]
└─$ nc -zv 192.168.1.1 1433
192.168.1.1: inverse host lookup failed: Host name lookup failure
(UNKNOWN) [192.168.1.1] 1433 (ms-sql-s) : Connection timed out
```

Service Access Control Testsing

Test 3: Rate Limiting Effectiveness LAN → DMZ (ssh rate limiting)

```
user@user:~$ for i in {1..10}; do echo "Attempt $i"; timeout 5 ssh -p 22 -o ConnectTimeout=5 testuser@10.0.1.10; sleep 2; done
Attempt 1
testuser@10.0.1.10's password:
Attempt 2
Attempt 3
testuser@10.0.1.10's password:
Attempt 4
Attempt 5
Attempt 6
testuser@10.0.1.10's password:
Attempt 7
Attempt 8
testuser@10.0.1.10's password:
Attempt 9
Attempt 10
user@user:~$
```

NAT Functionality Testing

Test 1: LAN Internet Access via NAT

```
user@user:~$ ping -c 3 192.168.1.10
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.
64 bytes from 192.168.1.10: icmp_seq=1 ttl=63 time=20.3 ms
64 bytes from 192.168.1.10: icmp_seq=2 ttl=63 time=2.50 ms
64 bytes from 192.168.1.10: icmp_seq=3 ttl=63 time=13.7 ms

--- 192.168.1.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 2.502/12.189/20.317/7.356 ms
```

```
┌──(kali㉿kali)-[~]
└─$ python3 -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
192.168.1.1 - - [25/Feb/2026 22:54:43] "GET / HTTP/1.1" 200 -
```

On WAN VM running port 80 server – we should get NAT IP address (firewall's) log

LAN → WAN (should succeed)

```
user@user:~$ curl -v http://192.168.1.10
*   Trying 192.168.1.10:80...
* Connected to 192.168.1.10 (192.168.1.10) port 80
> GET / HTTP/1.1
> Host: 192.168.1.10
> User-Agent: curl/8.5.0
> Accept: */*
>
* HTTP 1.0, assume close after body
< HTTP/1.0 200 OK
< Server: SimpleHTTP/0.6 Python/3.13.11
< Date: Thu, 26 Feb 2026 03:54:43 GMT
< Content-type: text/html
< Content-Length: 257
< Last-Modified: Thu, 26 Feb 2026 03:52:50 GMT
<
<!DOCTYPE html>
<html>
<head>
    <title>WAN Web Server</title>
</head>
<body>
    <h1>WAN Web Server Test Page</h1>
    <p>Server: wan (attacker)</p>
    <p>IP Address: 192.168.1.10</p>
    <p>Timestamp: Tue Feb 24 04:56:36 PM GMT 2026</p>
</body>
</html>
* Closing connection
```

NAT Functionality Testing

Test 2: DMZ Internet Access (Limited)

```
user@dmz:~$ nc -zv 192.168.1.10 25
nc: connect to 192.168.1.10 port 25 (tcp) failed: Connection timed out
user@dmz:~$ nc -zv 192.168.1.10 110
nc: connect to 192.168.1.10 port 110 (tcp) failed: Connection timed out
user@dmz:~$ _
```

(other ports, that are not 80 should not allow)

DMZ → WAN (should succeed)

```
user@dmz:~$ curl -v http://192.168.1.10
*   Trying 192.168.1.10:80...
* Connected to 192.168.1.10 (192.168.1.10) port 80
> GET / HTTP/1.1
> Host: 192.168.1.10
> User-Agent: curl/8.5.0
> Accept: */*
>
* HTTP 1.0, assume close after body
< HTTP/1.0 200 OK
< Server: SimpleHTTP/0.6 Python/3.13.11
< Date: Thu, 26 Feb 2026 04:01:31 GMT
< Content-type: text/html
< Content-Length: 257
< Last-Modified: Thu, 26 Feb 2026 03:52:50 GMT
<
<!DOCTYPE html>
<html>
<head>
    <title>WAN Web Server</title>
</head>
<body>
    <h1>WAN Web Server Test Page</h1>
    <p>Server: wan (attacker)</p>
    <p>IP Address: 192.168.1.10</p>
    <p>Timestamp: Tue Feb 24 04:56:36 PM GMT 2026</p>
</body>
</html>
* Closing connection
```

Attack Resistance Testing

Test 1: Port Scanning Detection

WAN → DMZ

```
┌──(kali㉿kali)-[~]
└─$ nmap -sS -p- 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-27 16:28 -0500
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.1.1
Host is up (0.0086s latency).
Not shown: 65533 filtered tcp ports (no-response)
PORT    STATE  SERVICE
80/tcp  open   http
443/tcp closed https
MAC Address: 08:00:27:2F:7C:6C (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 331.05 seconds
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sN -p- 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-28 08:11 -0500
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.1.1
Host is up (0.00086s latency).
All 65535 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 65535 open|filtered tcp ports (no-response)
MAC Address: 08:00:27:2F:7C:6C (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2925.57 seconds
```

.1 LEN=40 TOS=0x00 PREC=0x00 TTL=37 ID=51482 PROTO=TCP SPT=42277 DPT=44313 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.562744+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=39 ID=2205 PROTO=TCP SPT=42277 DPT=33291 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.562745+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=38 ID=39317 PROTO=TCP SPT=42277 DPT=28885 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.562746+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=55 ID=64532 PROTO=TCP SPT=42277 DPT=31922 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.562748+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=46 ID=18300 PROTO=TCP SPT=42277 DPT=63673 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.814253+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=54 ID=49533 PROTO=TCP SPT=42279 DPT=63673 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.814354+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=53 ID=49942 PROTO=TCP SPT=42279 DPT=31922 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.814389+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=52 ID=56985 PROTO=TCP SPT=42279 DPT=28885 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.814439+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=47 ID=53046 PROTO=TCP SPT=42279 DPT=33291 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.814483+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=56 ID=19863 PROTO=TCP SPT=42279 DPT=44313 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.814515+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=40 ID=22648 PROTO=TCP SPT=42279 DPT=48524 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.814545+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=38 ID=34547 PROTO=TCP SPT=42279 DPT=23775 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.814574+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=56 ID=61981 PROTO=TCP SPT=42277 DPT=20006 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.814605+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=39 ID=25856 PROTO=TCP SPT=42277 DPT=5706 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.814635+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=43 ID=8067 PROTO=TCP SPT=42277 DPT=54716 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.906109+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=44 ID=51331 PROTO=TCP SPT=42279 DPT=54716 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.906204+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=56 ID=34813 PROTO=TCP SPT=42279 DPT=5706 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.906233+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=46 ID=37857 PROTO=TCP SPT=42279 DPT=20006 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.906326+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=48 ID=54202 PROTO=TCP SPT=42277 DPT=22542 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.906448+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=58 ID=38434 PROTO=TCP SPT=42277 DPT=31809 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.906557+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=43 ID=1705 PROTO=TCP SPT=42277 DPT=28893 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.906622+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=57 ID=3792 PROTO=TCP SPT=42277 DPT=36983 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.906713+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=46 ID=46489 PROTO=TCP SPT=42277 DPT=47044 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.906776+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=46 ID=37805 PROTO=TCP SPT=42277 DPT=19812 WINDOW=1024 RES=0x00 URGP=0
2026-02-28T13:11:19.907518+00:00 firewall kernel: WAN-TO-FW-BLOCKED: IN=enp0s3 OUT= MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=192.168.1
.1 LEN=40 TOS=0x00 PREC=0x00 TTL=51 ID=52973 PROTO=TCP SPT=42277 DPT=35895 WINDOW=1024 RES=0x00 URGP=0

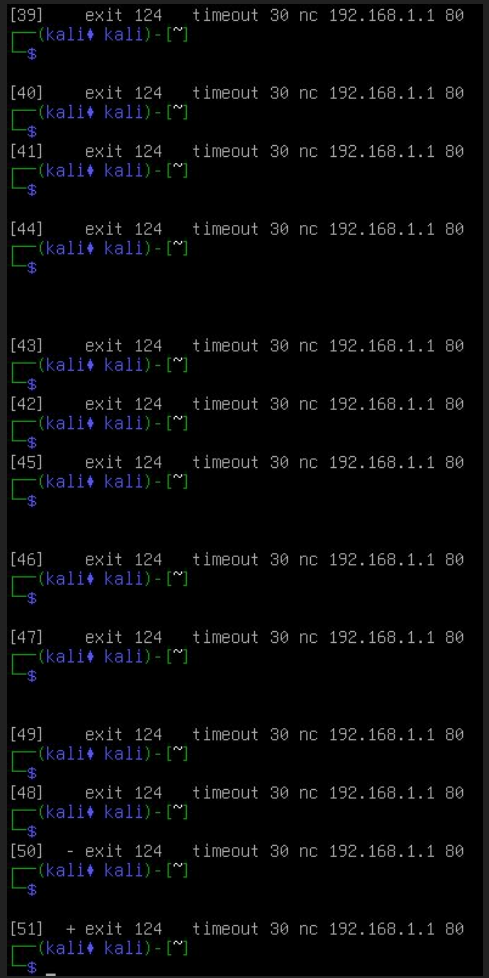
Attack Resistance Testing

Test 2: Connection Flooding Attack

WAN → DMZ (rate limiting)

Test 3: Anti-Spoofing Protection

WAN → DMZ (block & logging)



```
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=28682 PROTO=TCP SPT=2803 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692232+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=7219 PROTO=TCP SPT=2804 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692233+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=49260 PROTO=TCP SPT=2805 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692235+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=33636 PROTO=TCP SPT=2806 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692236+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=4592 PROTO=TCP SPT=2807 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692238+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=21605 PROTO=TCP SPT=2808 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692239+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=35541 PROTO=TCP SPT=2809 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692240+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=5562 PROTO=TCP SPT=2810 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692242+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=5424 PROTO=TCP SPT=2813 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692243+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=26318 PROTO=TCP SPT=2814 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692244+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=41339 PROTO=TCP SPT=2815 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692246+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=48084 PROTO=TCP SPT=2816 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692247+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=21220 PROTO=TCP SPT=2817 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692249+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=44049 PROTO=TCP SPT=2818 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692250+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=21250 PROTO=TCP SPT=2820 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692251+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=37719 PROTO=TCP SPT=2821 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692253+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=54808 PROTO=TCP SPT=2822 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692254+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=9530 PROTO=TCP SPT=2823 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.692256+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=31057 PROTO=TCP SPT=2825 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.749388+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=7492 PROTO=TCP SPT=2826 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.749413+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=9111 PROTO=TCP SPT=2827 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.749415+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=37031 PROTO=TCP SPT=2828 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.749417+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=32586 PROTO=TCP SPT=2829 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.749418+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=37811 PROTO=TCP SPT=2830 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
2026-02-26T15:49:11.749420+00:00 firewall kernel: SPOOFED-SRC: IN=enp0s3 OUT=enp0s8 MAC=08:00:27:2f:7c:6c:08:00:27:63:b0:05:08:00 SRC=192.168.1.10 DST=10.0.1.10
 LEN=40 TOS=0x00 PREC=0x00 TTL=63 ID=9855 PROTO=TCP SPT=2831 DPT=80 WINDOW=512 RES=0x00 SYN URGP=0
```

Administrative Access Testing

Test 1: SSH Administrative Access

LAN → DMZ (should succeed)

```
user@user:~$ ssh user@172.16.1.1
user@172.16.1.1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)
```

WAN → DMZ (should fail)

```
┌──(kali㉿kali)-[~]
└─$ ssh user@192.168.1.1
ssh: connect to host 192.168.1.1 port 22: Connection timed out
```

Firewall Comparison

*Conclusion

General comparison

Aspect	NFTables	pfSense	OPNsense
Learning curve	Steep	Moderate	Easy
Type	Linux packet filtering framework default in many distributions	Firewall/router OS distribution	Firewall/router OS distribution
Base system	Linux kernel	FreeBSD-based	FreeBSD-based
Management	CLI	GUI/CLI	GUI/CLI
Performance	Extremely high	Good	Good
Community	Community	Enterprise & community (has commercial version)	Community & enterprise
Hardware requirements	Minimal	Moderate	Moderate
Usage	Linux servers, cloud infrastructure, kubernetes	Offices, public wifi, small institutions, high security labs	Offices, public wifi, small institutions, high security labs

Test-based comparison

- Mostly identical
- Differ in reconnaissance response (nmap -sS SYN packets)

```
┌──(kali㉿kali)-[~]
└─$ nmap -sS -p 1-1000 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-27 07:31 -0500
Nmap scan report for 192.168.1.1
Host is up (0.00054s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 08:00:27:B8:1B:C1 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 26.69 seconds
```

Virusprot Table

IP Address
192.168.1.10

```
┌──(kali㉿kali)-[~]
└─$ nmap -sS -p 80 192.168.1.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-26 16:41 EST
Nmap scan report for 192.168.1.1
Host is up (0.0011s latency).

PORT   STATE SERVICE
80/tcp open  http
MAC Address: 08:00:27:4E:BF:DA (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 13.20 seconds
```

Nftables	pfSense	OPNsense
Does not hide configuration	Hides configuration by temporary source IP block	Does not hide configuration

References

1. Netfilter.org (2024) nftables wiki. [Online] Available at: https://wiki.nftables.org/ [Accessed 25 February 2026].
2. Netfilter.org (2023) Man page of NFT. [Online] Available at: https://www.netfilter.org/projects/nftables/manpage.html [Accessed 25 February 2026].
3. Netgate (2026) pfSense Documentation. Available at: https://docs.netgate.com/pfsense/en/latest/ (Accessed: 25 February 2026).
4. OPNsense (2024) OPNsense Documentation. Available at: https://docs.opnsense.org (Accessed: 25 February 2026).
5. Deciso BV (2024) OPNsense: Open Source Firewall. Available at:https://docs.opnsense.org/firewall.html (Accessed: 25 February 2026).

Thanks for attention

any questions?